The empirical recurrence for OEIS A206211, proved by transfer matrix

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Abstract

OEIS A206211 counts, up to renaming the letters, the arrays of width 5 over at most 3 letters in which every 2×3 and 3×2 subblock has exactly 2 equal edges, an edge of such a window being one of the six steps of its perimeter cycle and an equal edge one whose two cells agree. It carries an empirical recurrence of order 3 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. Counting up to renaming is a fixed rational combination of ordinary array counts, each of those is a walk count, and the conjectured recurrence is then settled by a finite exact computation.

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1 The conjecture

OEIS A206211 is "Number of (n+1)X5 0..2 arrays with every 2X3 or 3X2 subblock having exactly two equal edges, and new values 0..2 introduced in row major order.". It has offset 1 and begins

$$594, 992, 2224, 8136, 21984, 74080, 274320, 747456, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 34*a(n-3) for n>7

As of the "Last modified" line on the live entry (Oct 03 2025 12:47:35, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The condition is that every 2×3 and 3×2 subblock has exactly 2 equal edges, an edge of such a window being one of the six steps of its perimeter cycle and an equal edge one whose two cells agree.

The clause "new values 0..2 introduced in row major order" says the array is the canonical representative of its EQUALITY PATTERN, so what is counted is patterns, not arrays. Write N_j for the number of admissible patterns using exactly j letters and L_i for the number of admissible arrays over an alphabet of i letters. An array over i letters is a pattern together with an injection of its letters into the alphabet, so

$$L_i \ = \ \sum_j N_j \, i(i-1)\cdots(i-j+1),$$

a triangular system with unit diagonal, hence invertible over $\mathbf{Q}$. The entry counts $a = N_1 + \cdots + N_3$, so

$$a \;=\; \sum_{i=1}^{3} w_i\, L_i$$

for one fixed rational vector $w = \mathbf{1}^\top F^{-1}$, F the matrix of falling factorials.

3 Why the reduction is allowed here

The step above is legitimate only if the condition is invariant under permuting the alphabet, and in this family that is a real restriction rather than a formality. An *equal edge* is an edge of a subblock whose two cells carry the same letter; whether two cells agree is untouched by any renaming, so every condition built from counting equal edges is invariant and the reduction applies.

The neighbouring conditions of the same family that count *increases* — how many steps of a subblock's perimeter cycle go UP in value — are not invariant: they compare letters by size, and a permutation of the alphabet changes the count. Those entries are therefore excluded here rather than reduced, and are settled, when they are settled, over their own alphabet without the pattern step.

4 Each L_i is a walk count

For a fixed alphabet the condition is local: a 2×2 subblock is built from two consecutive array rows, a 2×3 or 3×2 window from two or three of them, and every comparison the entry makes is between subblocks that meet in the grid. Taking as state a single row where the condition looks at one subblock at a time, and the pair of consecutive rows where it compares neighbours, each L_i is a walk count on its own digraph. Assembling,

$$a(n) \;=\; \iota^\top M^{\,j} \tau$$

on the disjoint union of those digraphs for $i = 1, \ldots, 3$, with ι carrying the weights w_i cleared of denominators and $S = 3698$ states in all. The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

5 The proof

Lemma 1. *Let $q(t) = t^3 - \sum_i c_i t^{3-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+3} - \sum_i c_i M^{j+3-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; 34\, a(n-3)$$

for every $n > 7$.

Proof. The vector $w = q(M)\tau$ was formed by 3 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots$, again exactly; those integers vanish from the index corresponding to $n = 7 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. The rational weights are cleared by a common denominator first, so the whole computation is over the integers. The bound is exact: at $n = 7$ the recurrence fails on the entry's own published terms, so no larger range holds. $\square$

6 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 22 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A206211.

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